

LAB - In class

Topic 2: Simple SQL Queries



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In-class assignment

Task 1:

- Run the given UniversityDB_script file to create the University database and import its data.

Task 2:

Utilize the SQL editor of SQL Management Studio or Azure Data Studio to write the following queries:

1. Find the Students who were born between 2000s and 2005s.
2. Find the students enrolled in the course 'CS07' offered in the Fall semester of 2022.
3. Extract the names and salary (after a 10% increase) of the instructors teaching the course CS07. (Note: Students insert appropriate sample data to test the query)
4. Retrieve the (course_id, course_Name, prerequisite name) for all courses in the 'IS' department.
5. Retrieve the instructors of department 'Computer Science' whose salary is between \$2000 and \$3000.
6. Retrieve the instructors and their teaching courses with the role "lecturer". The result should be sorted first by department name and then alphabetically by instructor name.
7. Extract a list of (course id, course name) where instructor 'I002' has not participated in teaching.
8. Provide a list of instructors (instructor_id) who have participated in teaching in both the role of 'lecturer' and the role of 'TA'.
9. Retrieve a list of (student_id, student_name) who passed the course CS03 but failed the course CS04 in the Fall semester of 2022.
10. Retrieve the codes and names of students who have passed more than two courses.
11. Retrieve the school years and semesters, along with the number of courses held in each semester of every year.
12. Retrieve the codes and names of courses with the highest number of instructors who have taught those courses.
13. Retrieve the codes and names of instructors who have only taught courses managed by their own department.
14. Retrieve the code and name of **ALL** students, along with the count of courses managed by each department that they have participated in.
15. Find the top scoring student in each section. Print all top students with their IDs if their scores are tied.